

Making Magic Happen: Working with JupyterHub and Github

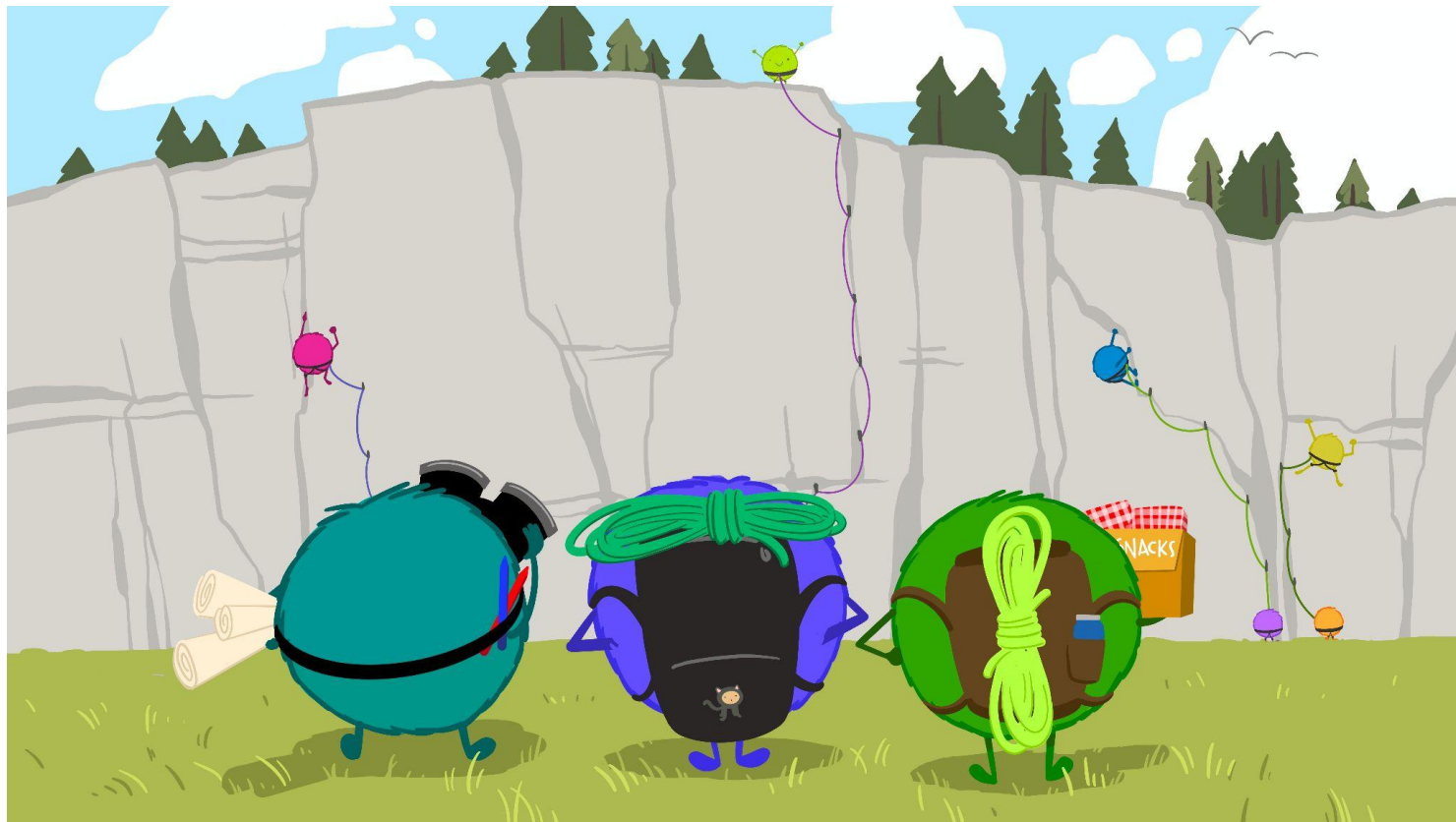


Why JupyterHub & Github?

- ❑ Browser Access
- ❑ Has R, Python pre-loaded
- ❑ Has Data pre-loaded
- ❑ One Stop Shop (data, computer)
- ❑ Work Together, but Separate
- ❑ Warning: Clone v Fork



Let's Do This Together!



What you need to get started:

Github Account: <https://github.com/signup>

Share your Github username

Check Your Email for an Invitation
to our Repo– reach out to us if
you don't have an invite!

Generate Github Personal Token

We are going to generate a **classic** token.

1. Go to <https://github.com/settings/tokens>
2. Click Generate new token > Generate new token (classic)
3. When the pop-up shows up, fill in a note, click the “repo” checkbox under “Select scopes”, and then scroll to bottom to click “Generate”
4. SAVE the token. You'll need it when working between JupyterHub and Github

(you can access it here: <https://github.com/settings/tokens>)



Log into JupyterHub

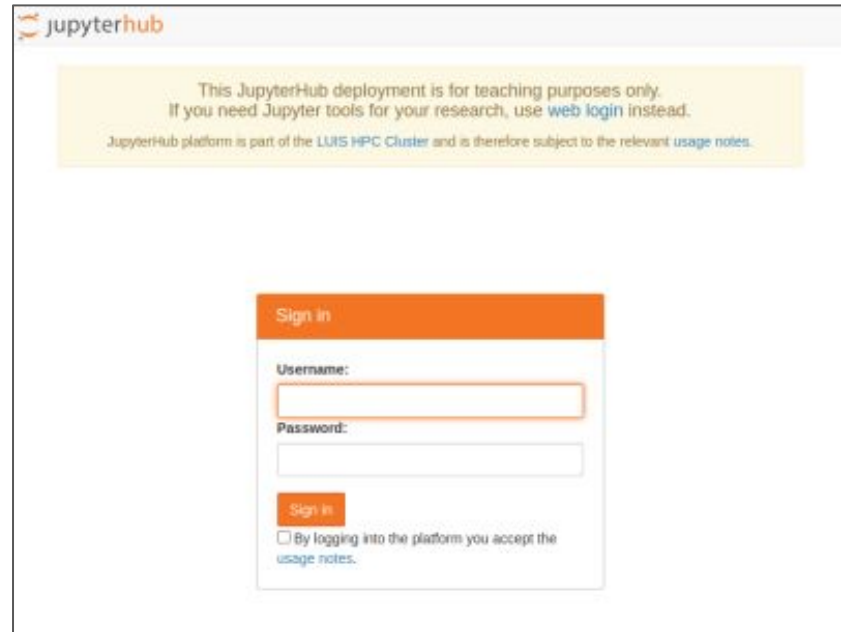
<https://workshop.nmfs-openscapes.2i2c.cloud>

Username: your email

Password: rodeo4UWrobots@2026

(or cudem2026 if that doesn't work!)

Use default image and resource
allocation settings for now!



The screenshot shows the JupyterHub login interface. At the top, the JupyterHub logo is visible. Below it, a yellow banner contains the following text: "This JupyterHub deployment is for teaching purposes only. If you need Jupyter tools for your research, use [web login](#) instead. JupyterHub platform is part of the [LUTS HPC Cluster](#) and is therefore subject to the relevant [usage notes](#)." Below the banner is a "Sign in" form with an orange header. The form includes fields for "Username:" and "Password:", a "Sign in" button, and a checkbox labeled "By logging into the platform you accept the usage notes."

Use Default Selections (for now)

Server Options

Default

[Permalink](#)

Choose image and resource allocation

Image	Py-R - Geospatial + QGIS, Panoply - py-rocket-geospatial-2 2026.02.18
Resource Allocation	~2 GB RAM, ~0.2 CPUs Up to ~4 CPUs when available

Start

If these are now powerful enough (you start to use all your memory), you can stop the server & restart, using more ram.

Navigating JupyterHub

The screenshot displays the JupyterLab user interface. A red rectangular box highlights the left-hand sidebar, specifically the file browser panel. This panel shows the current directory as `/ shared-public / GliderRodeo /`. Below the directory path, there is a table listing files and folders with their names and modification times.

Name	Modified
belladonna_20260...	22h ago
capex987_20260128	22h ago
Noise	9m ago
risso_20260128	22h ago
SEA117-M026_20...	22h ago
sg274_20260128	22h ago
sg607_20260128	22h ago
stenella_20260128	23h ago
Tutorials	20h ago
PAM_Glider_Specs...	22h ago
Slocum Glider Ro...	22h ago
waypoints.csv	22h ago

The main area of the interface shows the path `shared-public/GliderRodeo` and offers various tool options categorized into:

- Notebook:** Includes Python 3 (ipykernel), Desktop, R 4.5.1, RStudio, and VS Code.
- Console:** Includes Python 3 (ipykernel) and R 4.5.1.
- Other:** Includes Terminal, Text File, Markdown File, Python File, R File, and Show Contextual Help.
- Jupytertext:** Includes Python - Percent Format, R - Percent Format, and MyST Markdown.

The bottom status bar indicates the interface is in 'Simple' mode and shows memory usage as 'Mem: 243.75 / 1861.02 MB'.

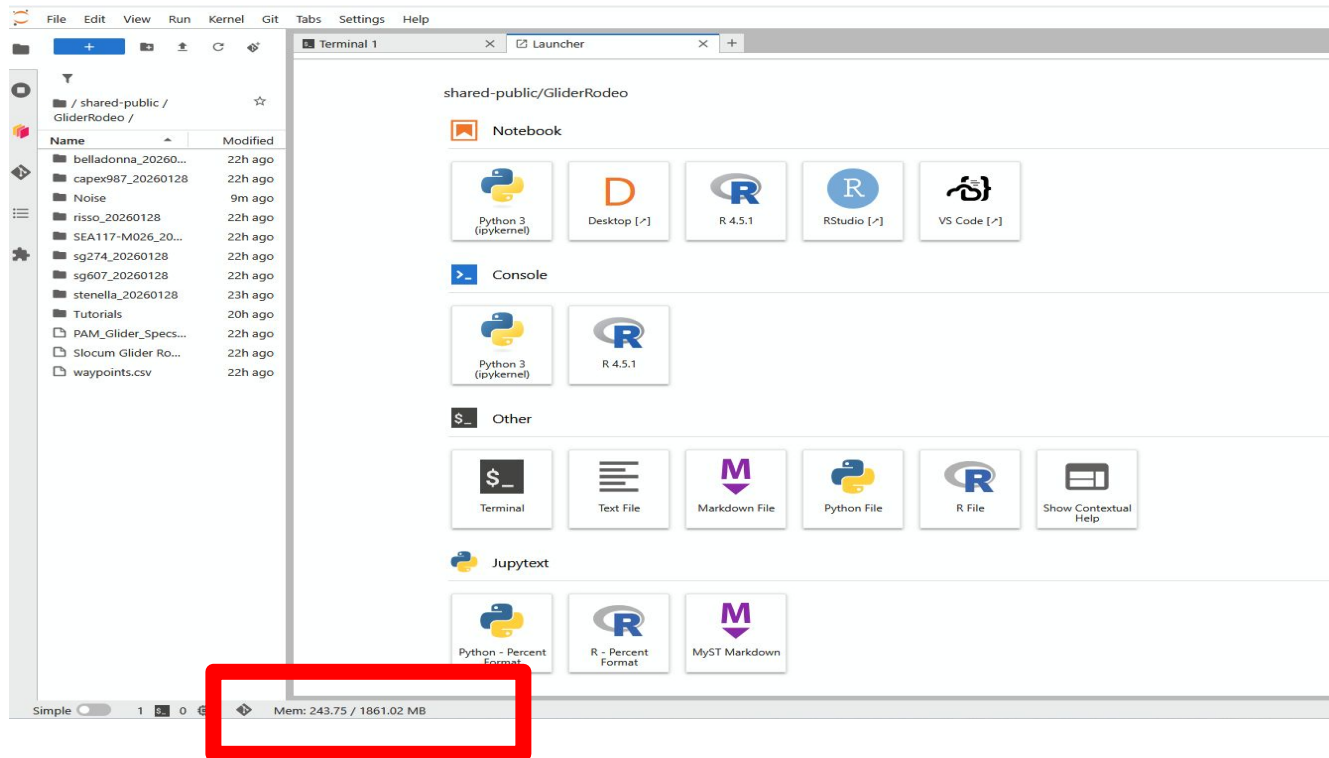
Navigating JupyterHub

The screenshot displays the JupyterHub web interface. At the top, a navigation bar includes links for Run, Kernel, Git, Tabs, Settings, and Help. Below this, a tab bar shows 'Terminal 1' and 'Launcher'. The main content area is titled 'shared-public/GliderRodeo' and is divided into several sections:

- Notebook:** Contains icons for Python 3 (ipykernel), Desktop [↗], R 4.5.1, RStudio [↗], and VS Code [↗].
- Console:** Contains icons for Python 3 (ipykernel) and R 4.5.1.
- Other:** Contains icons for Terminal, Text File, Markdown File, Python File, R File, and Show Contextual Help.
- Jupytertext:** Contains icons for Python - Percent Format, R - Percent Format, and MyST Markdown.

On the left side, a file browser shows a directory structure under '/ shared-public / GliderRodeo /'. A red box highlights a blue button with a white plus sign in the top-left corner of the file browser. The bottom status bar indicates 'Simple' mode, a memory usage of 'Mem: 243.75 / 1861.02 MB', and a small icon of a person.

Navigating JupyterHub



Configure Basic git Information

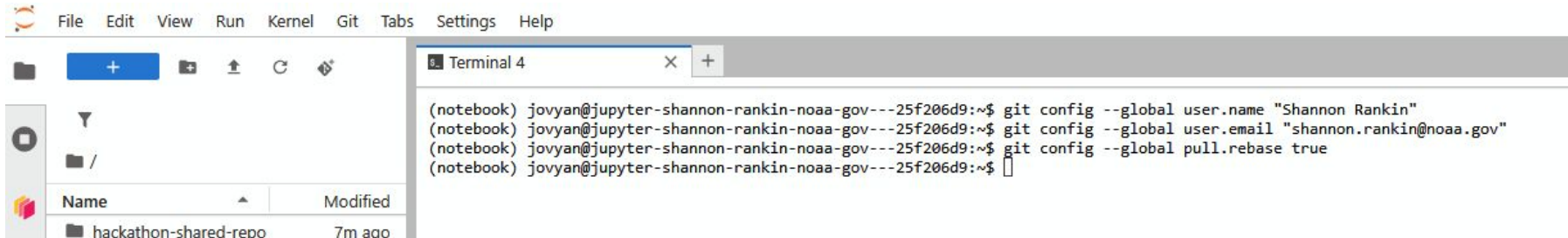
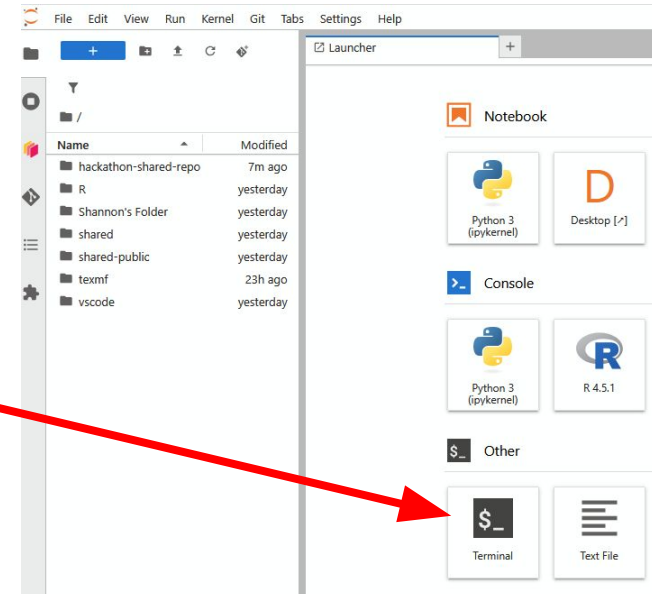
Open Terminal

```
$ git config --global user.name "Attendee Name"
```

```
$ git config --global user.email "attendee@hackweek.com"
```

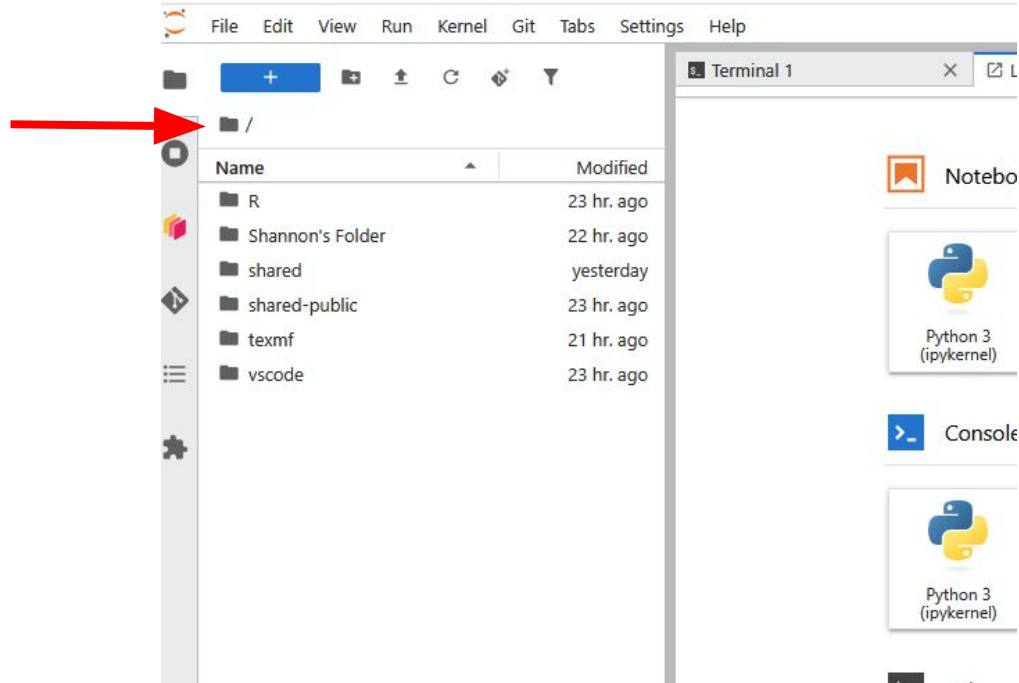
```
$ git config --global pull.rebase true
```

(but with YOUR Name and email)

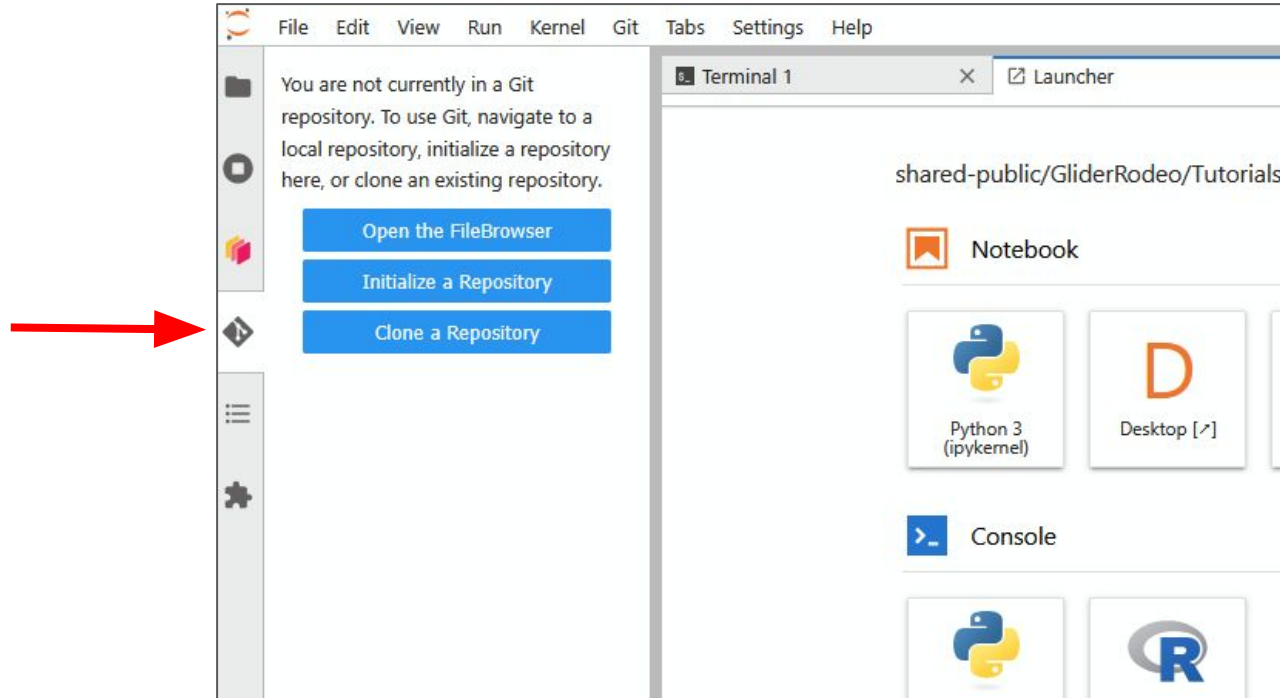


Clone Github Repository in JupyterHub

First, navigate up a folder (outside shared-public)

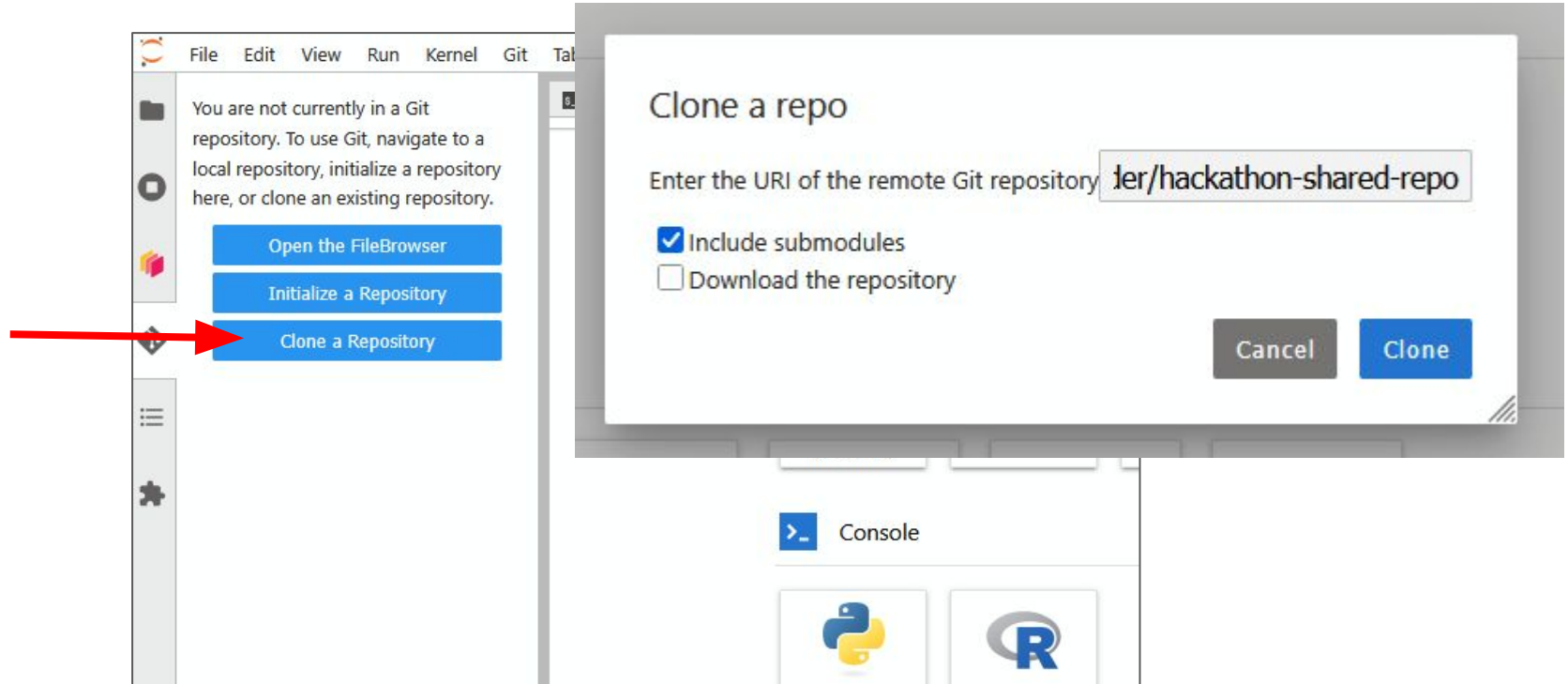


Clone Github Repository in JupyterHub



<https://github.com/NMFS-PAM-Glider/hackathon-shared-repo>

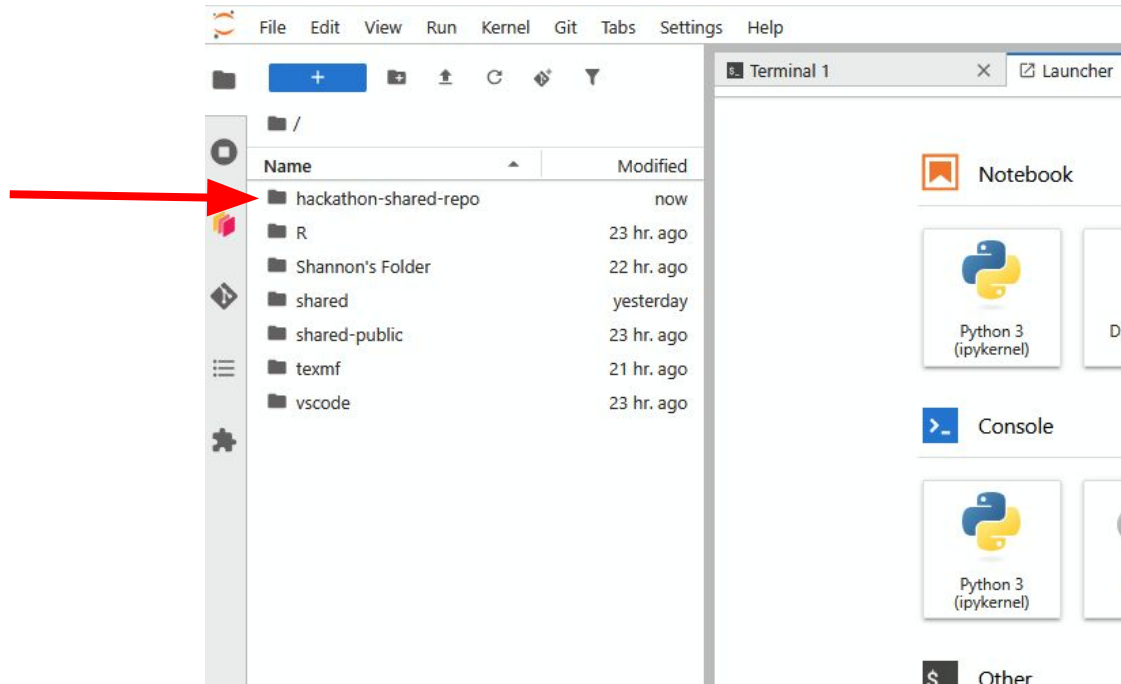
Clone Github Repository in JupyterHub



<https://github.com/NMFS-PAM-Glider/hackathon-shared-repo.git>

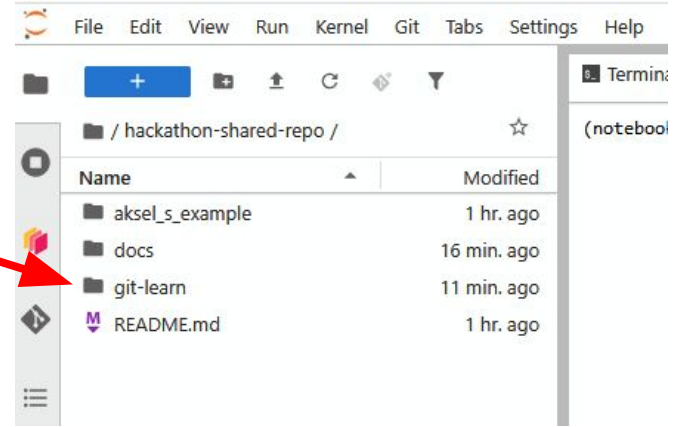
Clone Github Repository in JupyterHub

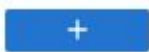
Now you can see it in your directory

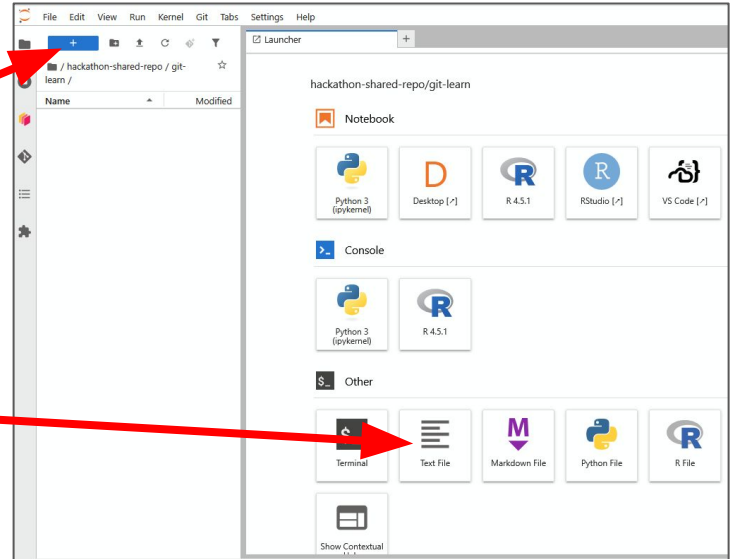


Add a File to git-learn folder

1. Navigate to git-learn folder;
2. Double-click to open folder



3. Click  to open Launcher

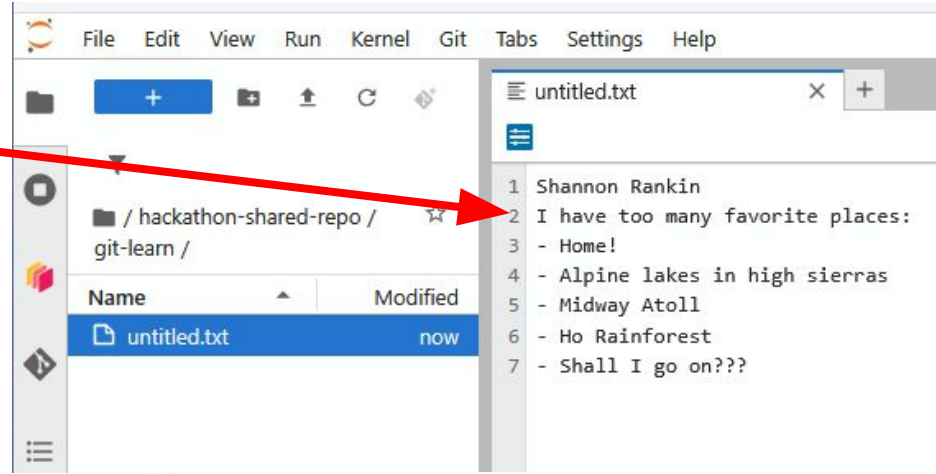


4. Open a text file

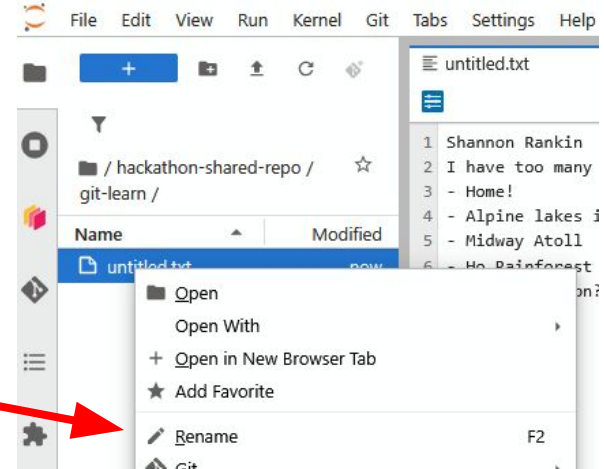
Add a File to git-learn folder

5. Write down your name and a favorite place or let us know your Superpower!

(or your favorite place)



6. Rename the text file to your (full) name. (or, if you type Control+S to save, it will ask you to name the file)



Save Changes to Github

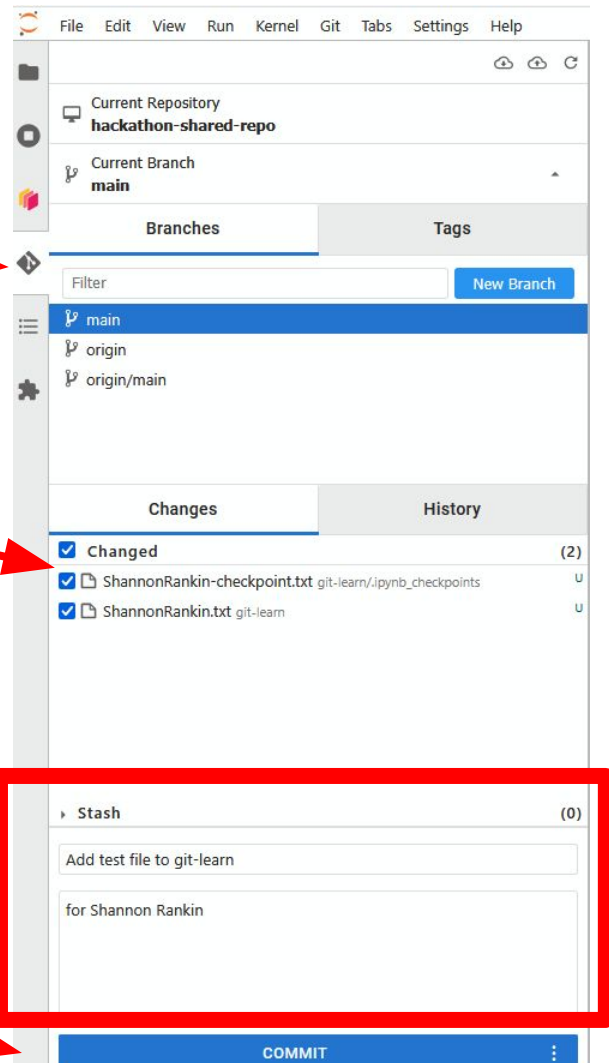
7. Navigate to Github again

8. Click on your changed files you want to commit

9. Give a 'summary' and description of the work you did (these are notes for future you (and your team!))

10. Click on

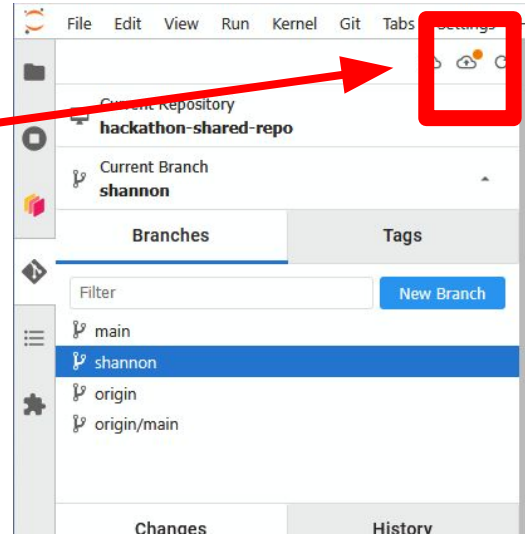
COMMIT



Push to Github

11. Push to Github

12. When prompted to enter your GitHub credentials, enter your username and token you made earlier



Check the Github Repo to see if your changes saved

Repo:

<https://github.com/NMFS-PAM-Glider/hackathon-shared-repo/>

Git-learn folder:

<https://github.com/NMFS-PAM-Glider/hackathon-shared-repo/tree/main/git-learn>

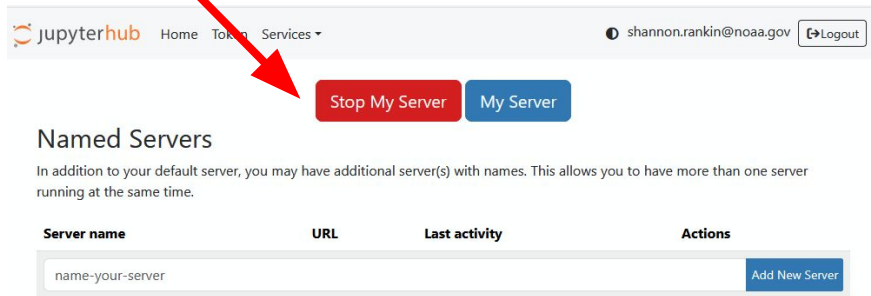
Log Out Properly

Go to File → Hub Control Panel
(near the bottom)

This will open the Hub Control

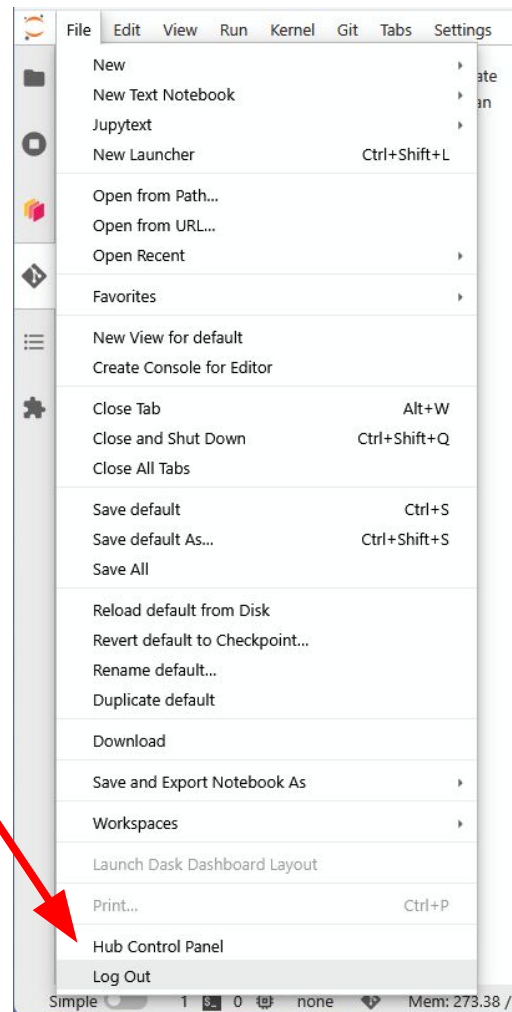
Click on

Stop My Server



The screenshot shows the JupyterHub web interface. At the top, there's a navigation bar with 'jupyterhub', 'Home', 'Token', and 'Services'. On the right, it shows the user 'shannon.rankin@noaa.gov' and a 'Logout' button. Below the navigation bar, there are two buttons: 'Stop My Server' (highlighted with a red arrow) and 'My Server'. Underneath, there's a section titled 'Named Servers' with a paragraph explaining that you can have multiple servers with names. Below this is a table with columns: 'Server name', 'URL', 'Last activity', and 'Actions'. The first row has 'name-your-server' in the 'Server name' column and an 'Add New Server' button in the 'Actions' column.

Server name	URL	Last activity	Actions
name-your-server			<button>Add New Server</button>

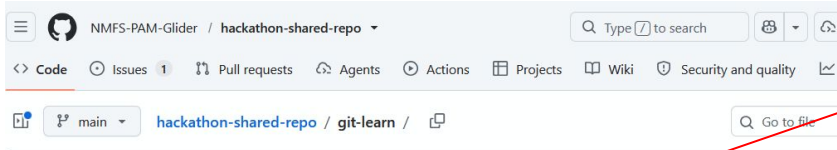


The screenshot shows the JupyterLab application window with the 'File' menu open. The menu items include: 'New', 'New Text Notebook', 'Jupyter', 'New Launcher' (with keyboard shortcut 'Ctrl+Shift+L'), 'Open from Path...', 'Open from URL...', 'Open Recent', 'Favorites', 'New View for default', 'Create Console for Editor', 'Close Tab' (with keyboard shortcut 'Alt+W'), 'Close and Shut Down' (with keyboard shortcut 'Ctrl+Shift+Q'), 'Close All Tabs', 'Save default' (with keyboard shortcut 'Ctrl+S'), 'Save default As...' (with keyboard shortcut 'Ctrl+Shift+S'), 'Save All', 'Reload default from Disk', 'Revert default to Checkpoint...', 'Rename default...', 'Duplicate default', 'Download', 'Save and Export Notebook As', 'Workspaces', 'Launch Dask Dashboard Layout', 'Print...' (with keyboard shortcut 'Ctrl+P'), 'Hub Control Panel' (highlighted with a red arrow), and 'Log Out'. The status bar at the bottom shows 'Simple', '1', '0', 'none', and 'Mem: 273.38 /'.

Check Github for your Revision

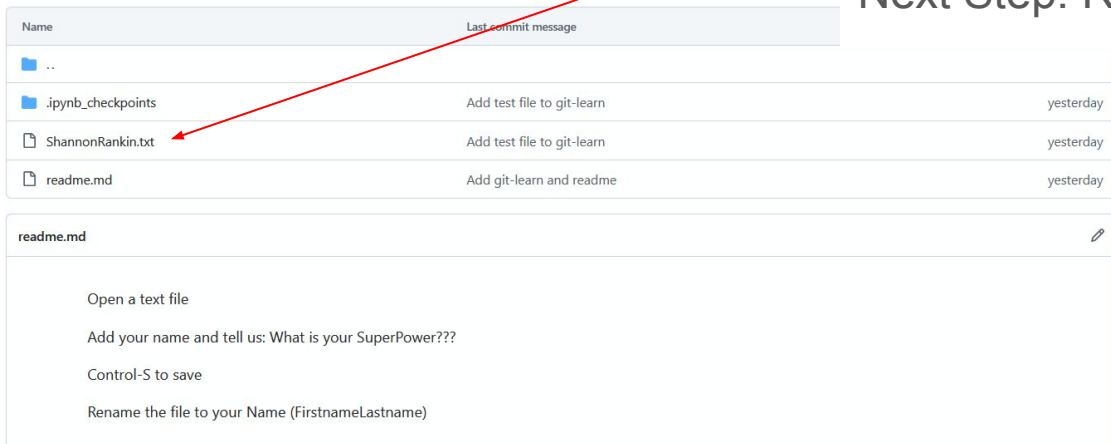
Go to git-learn folder on repository:

<https://github.com/NMFS-PAM-Glider/hackathon-shared-repo/tree/main/git-learn>



Give it a minute, do you see your file?

Next Step: Respond to Issue (see next page)



Github Issues

The screenshot displays the GitHub interface for a repository named 'hackathon-shared-repo'. The 'Issues' tab is selected and highlighted with a red box. The issue title is 'add text to git-learn #1'. The issue description states: 'As part of a learning exercise to work with Github in JupyterHub, each hackweek participant will follow a tutorial to add a text file to the git-learn folder in this repo. The filename will be their full name, and the text file will include their name and an anecdote about them. Here is a [link to the instructions](#). To get familiar with Issues, each participant should reply in a comment to [this Issue](#) and reply if they have completed the task, or if they are having problems (If they are having problems, then please state the problem and we will help you!).' The right sidebar shows settings for assignees, labels, type, fields, projects, milestone, relationships, and development. The bottom section shows a comment input area with a 'Close issue' button and a 'Comment' button.

Create a New Issue

Assign an Issue to a person in the Repo

Provide detailed information

Add updated info, Respond

When issue is completed, Close it.

Github Issues- Practice

1. [Go to Repo](#)
2. [Go to Issue #1](#)
3. Reply with a comment
 - a. You successfully added your document
 - b. If you have problems, state them here & assign to Shannon Rankin

The screenshot shows a GitHub issue page for the repository 'NMFS-PAM-Glider / hackathon-shared-repo'. The issue title is 'add text to git-learn #1'. The issue was opened 1m ago by shannonrankin. The issue description contains instructions for participants to add a text file to the 'git-learn' folder and to reply with a comment if they have problems. The issue is currently assigned to shannonrankin. The right sidebar shows the 'Assignees' section with shannonrankin listed, and the 'Labels' section with 'No labels'. The 'Fields' section shows 'Priority' as 'Choose an option'. The 'Projects' section shows 'No projects'. The 'Milestone' section shows 'No milestone'. The 'Relationships' section shows 'None yet'. The bottom section is titled 'Add a comment' and has a text area for writing a comment.

Lather, Rinse, Repeat

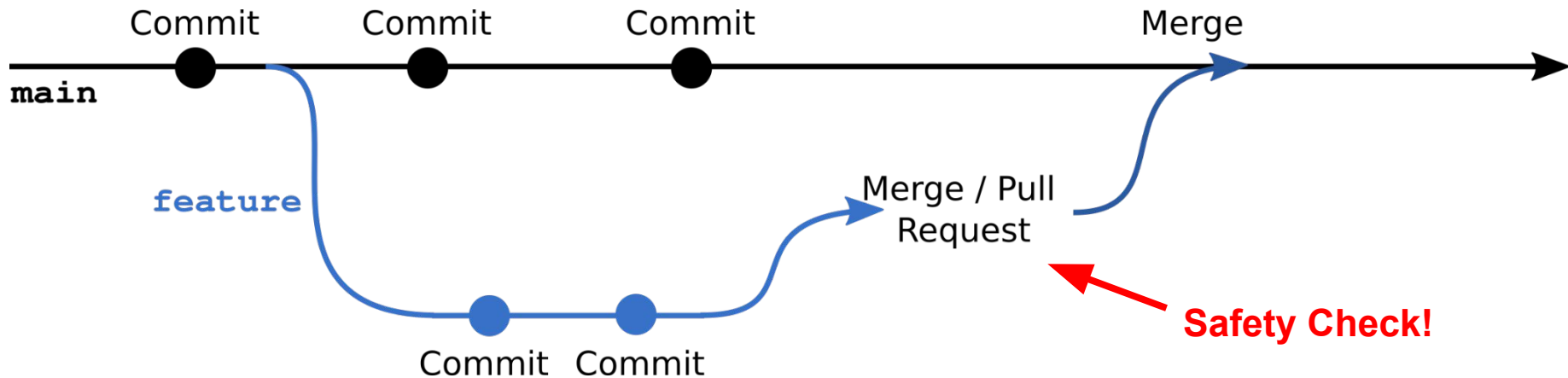
With each session, we'll repeat this general pattern:

- Log into JupyterHub
- Join (or update) Github Repo
- Do your work!
- Commit and push to Github Repo
- Check Repo to ensure it is there
- Log out of JupyterHub



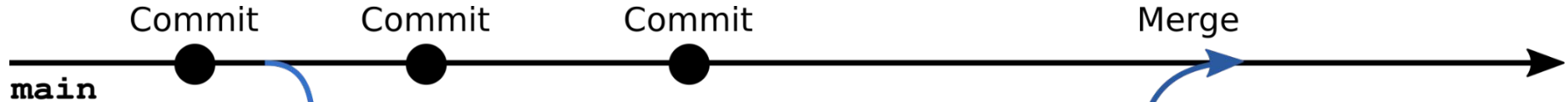


Warning!!





Warning!!



Safety Check!

- Only work in YOUR program Folder
- Segregate your work tasks
- If > 1 programmer → FORK!



Extra Slides

Prefer the Terminal over GUI?

task: Configure basic git information

command: \$ git config --global user.name "Attendee Name"
 \$ git config --global user.email "attendee@hackweek.com"
 \$ git config --global pull.rebase true
example: \$ git config --global user.name "John Doe"
 \$ git config --global user.email "john.doe@hackweek.com"
 \$ git config --global pull.rebase true

task: Verify your git configuration

command: \$ git config --list
example return: user.name=Attendee Name
 user.email=attendee@hackweek.com
 pull.rebase=true

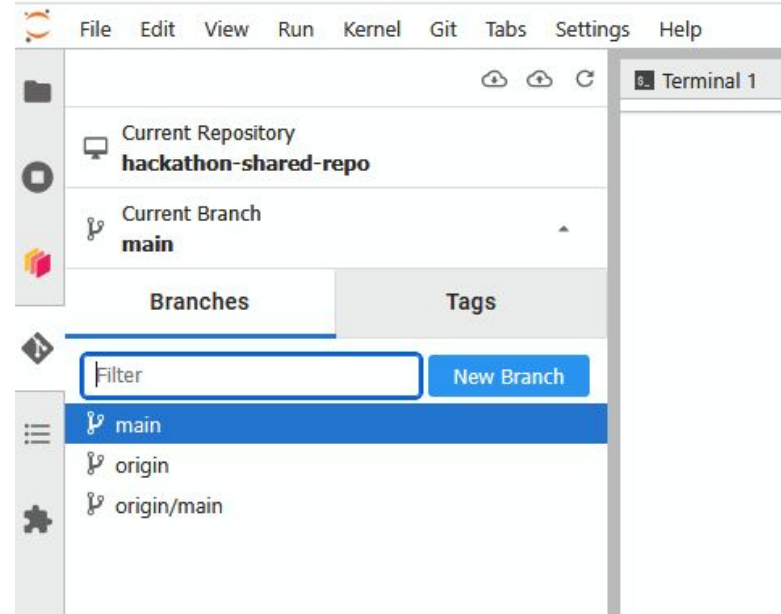
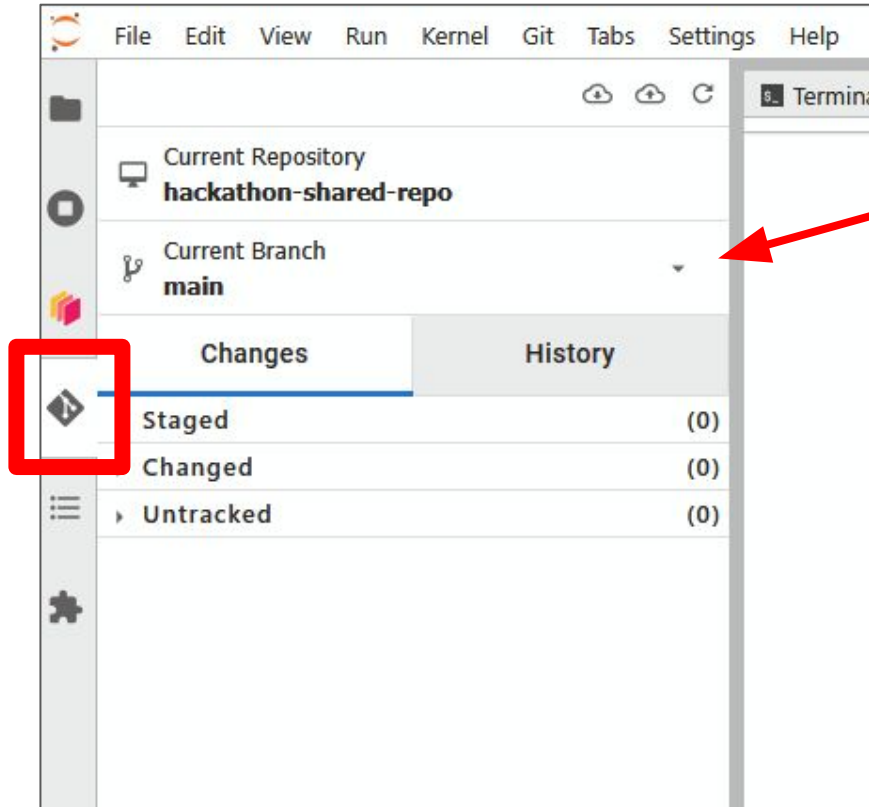
task: Clone a GitHub Repo

command: \$ git clone https://github.com/myRepo.git
example: \$ git clone https://github.com/NMFS-PAM-Glider/hackathon-shared-repo.git

task: Push Changes to a Repo (this will consist of a series of commands)

commands: \$ cd hackathon-shared-repo # Ensure you are in your intended directory
 \$ git status # Check on the status- you should have files to add
 \$ git add -A # to add all changed files
 \$ git commit -m "Summarize changes here"
 \$ git push

Create your own branch



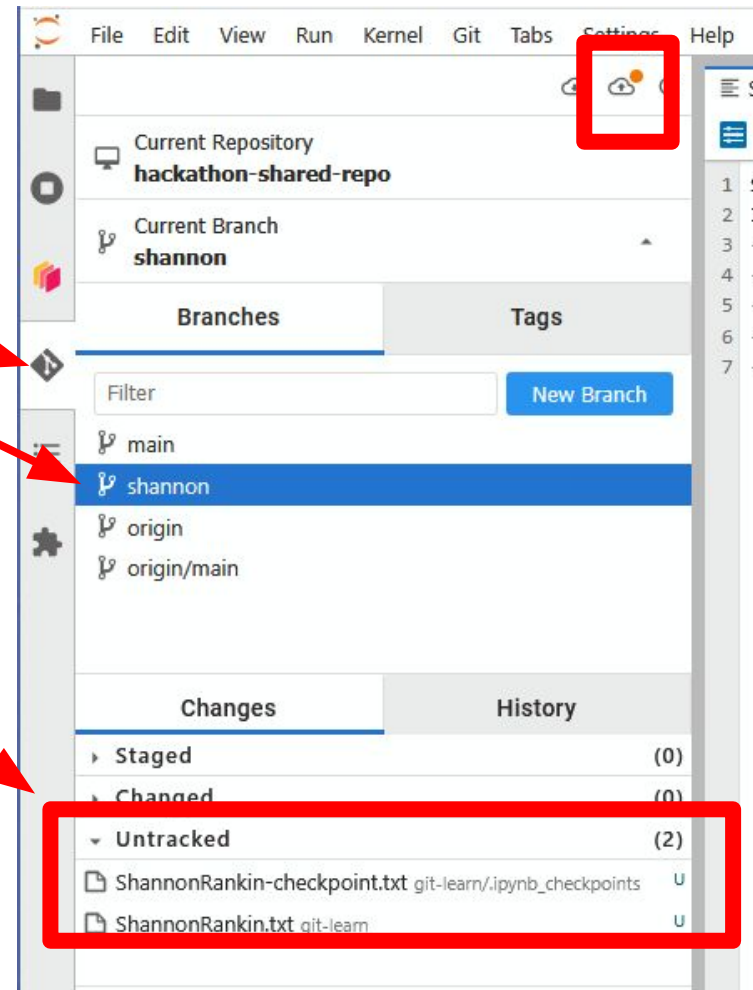
Save your Branch to Github

7. Navigate to Github again

8. Ensure you are in YOUR branch

9. Go to 'Untracked' and add to your branch by clicking on each item and selecting the '+' sign

(this will 'stage' them in your branch)



Save Changes to Github

7. Navigate to Github again

8. Ensure you are in YOUR branch

9. Go to 'Untracked' and add to your branch by clicking on each item and selecting the '+' sign

(this will 'stage' them in your branch)

